# Economic Growth, the Financial System, and Business Cycles

ECON 101 – Principles of Macroeconomics

Muhebullah Karimzada

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# Chapter Outline

- **6.1** Long-Run Economic Growth
- **6.2** Saving, Investment, and the Financial System
- **6.3** The Business Cycle

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# 6.1 Learning Objective

## Goal

Explain how long-run economic growth is measured and what determines it.

- **Long-run economic growth:** the process by which rising productivity increases the **average standard of living**.
- This differs from short-run fluctuations (**business cycles**):
  - **Expansion:** real GDP rising.
  - **Recession:** real GDP falling.
- The most common measure of the standard of living:
  - **Real GDP per capita** = real GDP ÷ population.

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# Real GDP per Capita: A Key Indicator

- **Real GDP per capita** tells us:
  - How much goods and services, on average, each person can consume.
  - Adjusted for changes in the price level (**real**, not nominal).
- Countries with higher real GDP per capita typically have:
  - better health, education, housing, and infrastructure.
- **Canadian context:** In 2023, Canada's real GDP per capita was approximately \$53,000 (2012 constant dollars), roughly **four times** the 1961 level of ≈\$13,800.
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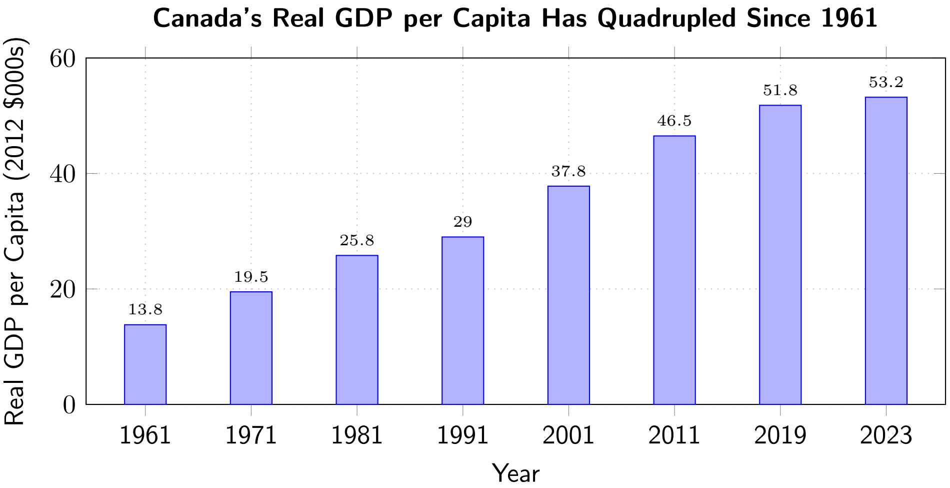
# Figure 6.1 – Real GDP per Capita in Canada, Selected Years




Source: Statistics Canada, Table 36-10-0222-01 (2012 constant dollars).

# Interpretation of Figure 6.1

- Measured in 2012 dollars, real GDP per capita has risen **nearly four-fold** since 1961.
- The average Canadian in 2023 commands almost four times as many goods and services as the average Canadian in 1961.
- This long-run rise is what we call **economic growth**.
- **Notable interruptions:**
  - Early 1980s recession (high interest rates, oil price shock).
  - Early 1990s recession (tight monetary policy, housing bust).
  - 2008–2009 Global Financial Crisis.
  - 2020 COVID-19 pandemic: real GDP per capita fell ≈13% in just 2 quarters — the sharpest decline on record.

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# Calculating Growth Rates

- Growth rate of an economic variable from year $t-1$ to year $t$:

$$\text{Growth rate}_t = \frac{\text{Value}_t - \text{Value}_{t-1}}{\text{Value}_{t-1}} \times 100.$$

- For a few years, compute growth each year and take the average.
- **Canadian example:**
  - Real GDP per capita in 2022: \$53,100.
  - Real GDP per capita in 2023: \$53,200.
  - Growth rate $\approx \dfrac{53{,}200 - 53{,}100}{53{,}100} \times 100 \approx \mathbf{0.19\%}$.
  - Canada's 2023 growth was very weak — immigration drove population faster than real output.

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# The Rule of 70

- A handy shortcut: the **Rule of 70**.
- If a variable grows at rate $g\%$ per year, the number of years to **double** is approximately:

$$\text{Years to double} \approx \frac{70}{g}.$$

- **Canadian examples:**
  - Canada's historical average growth of real GDP per capita $\approx 2\%$:
  - Years to double $\approx 70/2 = 35$ years.
  - At only $1\%$ growth: $70/1 = 70$ years — twice as long!
- Small differences in growth rates **compound** dramatically over decades.

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# Rule of 70: Country Comparison

**Country A — 2% growth**

- Years to double: $70/2 = 35$
- Income after 35 years: **2x**
- Income after 70 years: **4x**

**Country B — 4% growth**

- Years to double: $70/4 = 17.5$
- Income after 35 years: **4x**
- Income after 70 years: **16x**

- **Real example:** South Korea's economy grew $\approx 7\%$ per year from 1960 to 2000, doubling every 10 years — raising per capita income from ≈\$1,000 to ≈\$20,000 (2000 US\$).
- Canada's steadier $\approx 2\%$ average represents sustained prosperity without boom-and-bust volatility.

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# What Determines Long-Run Growth?

- Long-run increases in real GDP per capita depend on **labour productivity**:

$$\text{Labour productivity} = \frac{\text{Real GDP}}{\text{Number of workers (or hours worked)}}.$$

- The average Canadian produces much more per hour than in 1961.
- Two key drivers:
  1. **Increases in capital per hour worked**
  2. **Technological change**
- **Canadian data (Statistics Canada):** Labour productivity grew an average of 1.4% per year from 2000–2023 — slower than the G7 average, raising policy concern.

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# Capital per Hour Worked

- **Physical capital:** machines, computers, tools, buildings, vehicles.
  - More capital per worker $\Rightarrow$ more output per hour.
- **Human capital:**
  - Accumulated knowledge and skills from education, training, experience.
- **Canadian example:**
  - A Canadian oil sands worker equipped with modern extraction technology and earth-moving equipment produces orders of magnitude more output per day than a 1961 worker with manual tools.
  - BC's tech sector: engineers with university degrees and high-powered workstations represent both physical and human capital investment.

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# Technological Change

- **Technological change:** improvements in methods of combining inputs into outputs.
- Examples:
  - New production processes (AI-assisted manufacturing).
  - Better management (lean production, just-in-time inventory).
  - New products (mRNA vaccines, electric vehicles).
- Workers produce more **with the same inputs** — an outward shift of the production frontier.
- **Canadian example:** Shopify (Ottawa) enables thousands of small businesses to reach global markets with the same team size — a pure technology-driven productivity gain.
- **Entrepreneurs** pioneer these new ways of combining labour, capital, and ideas.

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# Potential GDP

- **Potential GDP:** the level of real GDP attained when all firms produce at **capacity** —
  - "normal" hours, "normal" workforce, no emergency overtime.
- Also called **full-employment GDP** or **trend GDP**.
- Potential GDP grows when:
  - the labour force expands (e.g., through immigration),
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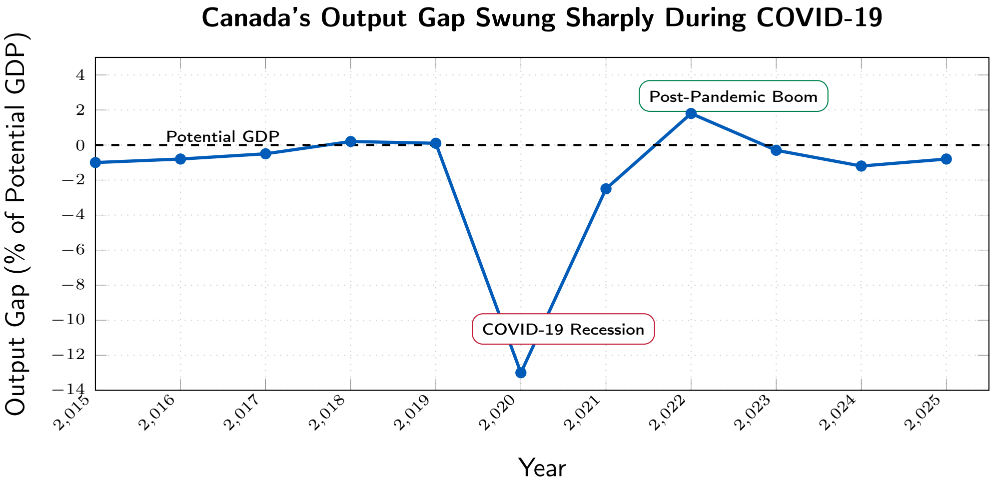
# Figure 6.2 – Output Gap: Canada 2015–2025 (Estimated)




Source: Bank of Canada, Monetary Policy Report (estimates). Shaded area = negative output gap (recessionary).

# The Output Gap

- **Output gap:** $=$ actual GDP $-$ potential GDP.
- When the output gap is **positive** (actual $>$ potential):
  - economy is using resources unsustainably — very low unemployment, high capacity utilization, **inflationary pressure**.
  - **Canada 2022**: output gap $\approx +1.8\%$ coincided with CPI inflation peaking at $8.1\%$.
- When the output gap is **negative** (actual $<$ potential):
  - resources (labour, capital) are underutilized.
  - **Canada 2020**: output gap plunged to $\approx -13\%$ during COVID-19 lockdowns.
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# 6.2 Learning Objective

## Goal

Explain how the financial system links saving and investment and why that matters for growth.

- Long-run growth requires:
  - more capital per worker, and ongoing technological improvements.
- Both require **investment**.
- Investment is financed by **saving**, channeled through the financial system.
- **Canadian saving rate** (household, % of disposable income): fell from ≈18% in 1982 to ≈2–4% pre-COVID, then surged to 28% during 2020 lockdowns (Statistics Canada).

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# An Overview of the Financial System

- **Financial markets:** where financial securities are bought and sold.
  - Toronto Stock Exchange (TSX), bond markets, money markets.
- **Stock:** financial security representing partial **ownership** of a firm.
  - Buying TSX-listed Shopify shares $\Rightarrow$ you own part of Shopify.
- **Bond:** financial security promising to repay a **fixed amount** of funds.
  - Government of Canada bonds, corporate bonds (e.g., TD Bank bonds).
  - A bond is effectively a **loan** from the buyer to the issuer.
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# Three Key Services of the Financial System

### Risk-Sharing

- Spread money across many assets.
- Reduces risk while maintaining returns.
- *E.g., RBC Select Balanced mutual fund.*

### Liquidity

- Ease of converting to cash.
- Bank accounts, money market funds, publicly traded stocks.
- Allows long-term commitments with short-term access.

### Information

- Prices of securities aggregate expectations.
- Guide funds toward the most productive uses.
- Reduce adverse selection and moral hazard.

## The Macroeconomics of Saving and Investment

- National income identity:

$$Y = C + I + G + NX.$$

- For simplicity, assume a **closed economy** ($NX = 0$):

$$Y = C + I + G \quad \Rightarrow \quad I = Y - C - G.$$

- Investment equals total income not consumed or used by government.
- **Private saving:**

$$S_{\text{private}} = Y + \text{TR} - C - T.$$

- **Public saving:**

$$S_{\text{public}} = T - G - \text{TR}.$$

- **Total saving = Investment:**

$$\boxed{S = S_{\text{private}} + S_{\text{public}} = I}$$

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- If $S_{\text{public}} < 0$: government runs a **budget deficit**.
  - Government must borrow (sell bonds), reducing funds available for private investment.
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- **Canadian context:**
  - Federal government ran deficits every year 2009–2024.
  - 2020–21 deficit: $314 billion (largest ever), due to COVID-19 emergency spending.
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- If $S_{\text{public}} < 0$: government runs a **budget deficit**.
    - Government must borrow (sell bonds), reducing funds available for private investment.
- **Crowding out:** the reduction in private investment caused by government borrowing.
- **Canadian context:**
    - Federal government ran deficits every year 2009–2024.
    - 2020–21 deficit: $314 billion (largest ever), due to COVID-19 emergency spending.
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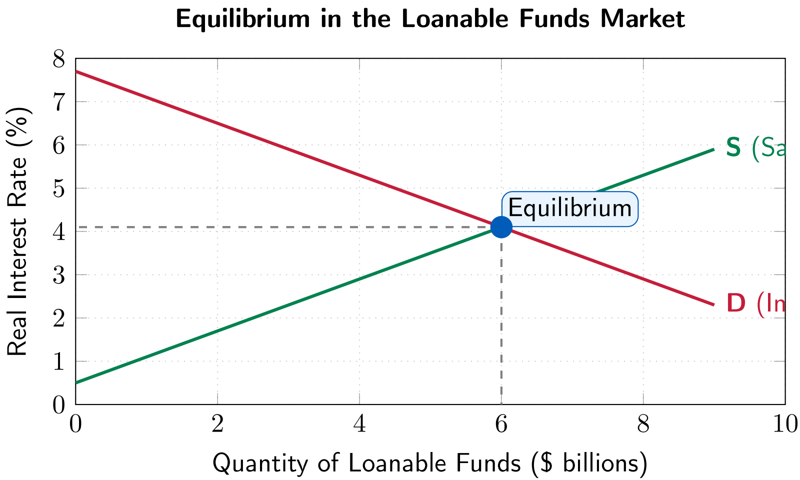
# Figure 6.3 – The Market for Loanable Funds



## Supply and Demand for Loanable Funds

- **Demand for loanable funds** (firms):
  - Firms borrow to finance investment projects.
  - Borrow more when real interest rate is **lower** (more projects profitable).
  - **Canada 2020**: pandemic uncertainty slashed private investment by ≈15%.
- **Supply of loanable funds** (households):
  - Households save more when real interest rate is **higher** (greater reward to saving).
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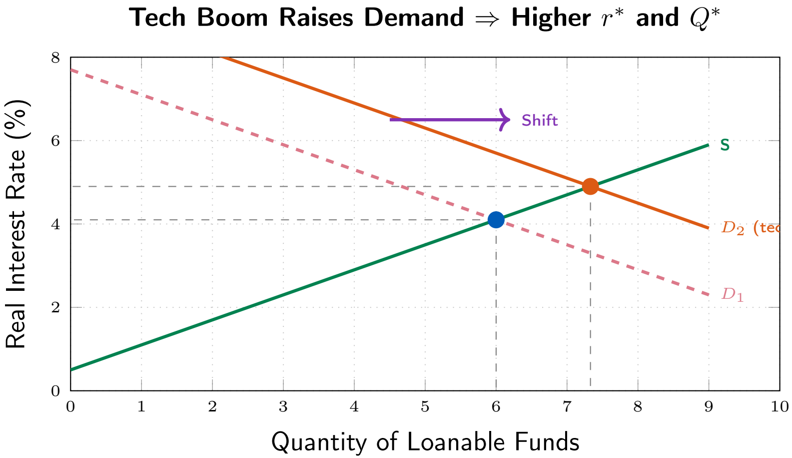
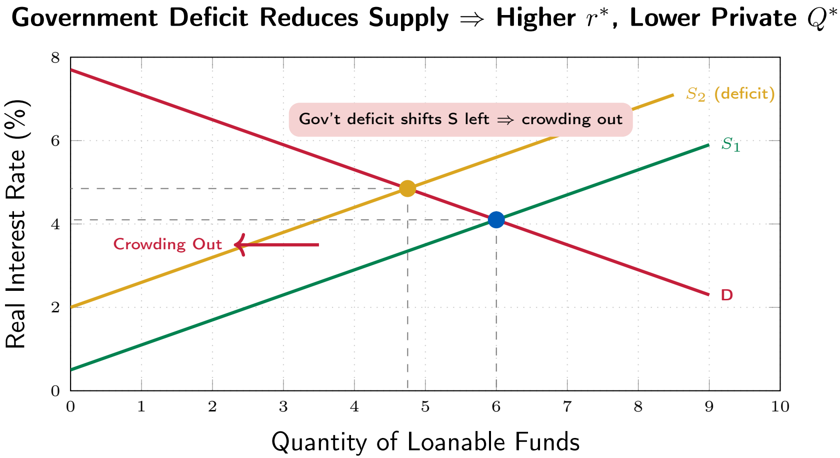


# Figure 6.5 – Budget Deficit Crowds Out Investment



# 6.3 Learning Objective

**Goal**

Explain what happens during the business cycle and how it affects inflation and unemployment.

- Real GDP per capita has risen about **four-fold** since 1961, but not smoothly.
- Canada has experienced several recessions:
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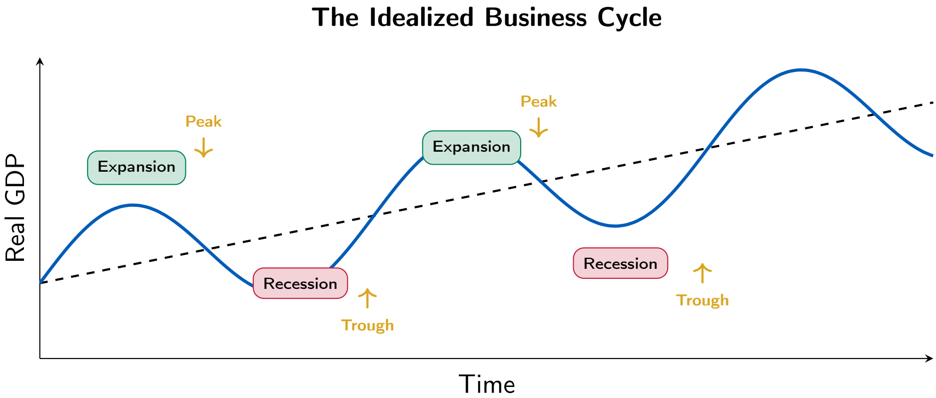
# Figure 6.6 – Phases of the Business Cycle



# Phases of the Business Cycle

- **Expansion:** real GDP is rising (employment grows, incomes rise).
- **Peak:** the point where real GDP stops rising and begins to fall.
- **Recession:** real GDP is falling (often defined as 2+ consecutive quarters).
- **Trough:** the point where real GDP stops falling and begins to rise.
- A new expansion follows the trough.

## Canadian Recessions Since 1980

| Recession | Cause | Peak Unemployment |
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| 1981–82 | Volcker rate shock | 13.0% |
| 1990–91 | Tight monetary policy | 11.9% |
| 2008–09 | Global financial crisis | 8.7% |
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- Common rule of thumb:
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- Statistics Canada (CANSIM) provides real GDP data quarterly and monthly.
- **2020 example:**
  - Q1 2020: real GDP fell −8.8% (annualized).
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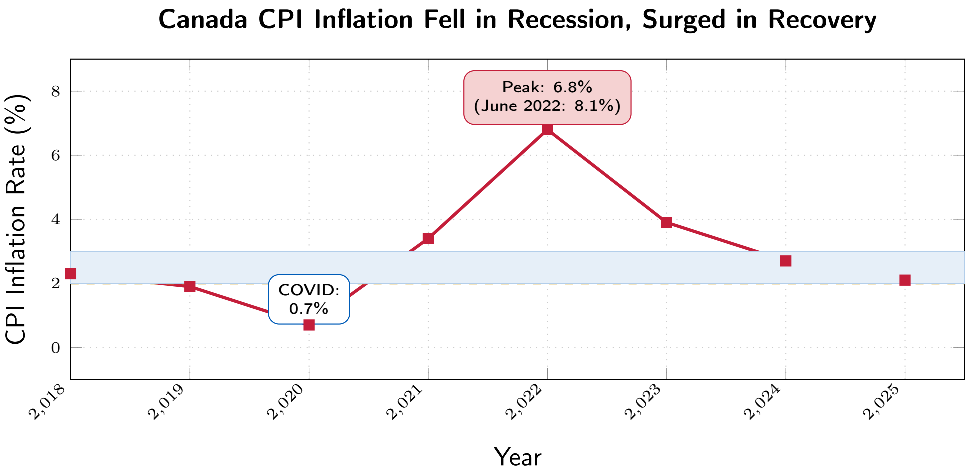
# Figure 6.7 – Inflation and the Business Cycle (Canada)




Source: Statistics Canada, Table 18-10-0004-01. 2025 is estimated. BoC target band 1–3% shaded.

# The Effect of the Business Cycle on Inflation

- **During recessions:**
  - Demand falls below supply capacity.
  - Firms compete for fewer customers $\Rightarrow$ prices rise more slowly or fall.
  - **Canada 2020**: CPI inflation dropped to 0.7%.
- **During expansions:**
  - Demand exceeds supply capacity.
  - Firms can raise prices more easily $\Rightarrow$ higher inflation.
  - **Canada 2022**: Post-COVID surge in demand + supply chain disruptions $\Rightarrow$ CPI peaked at 8.1% in June 2022 (Statistics Canada).
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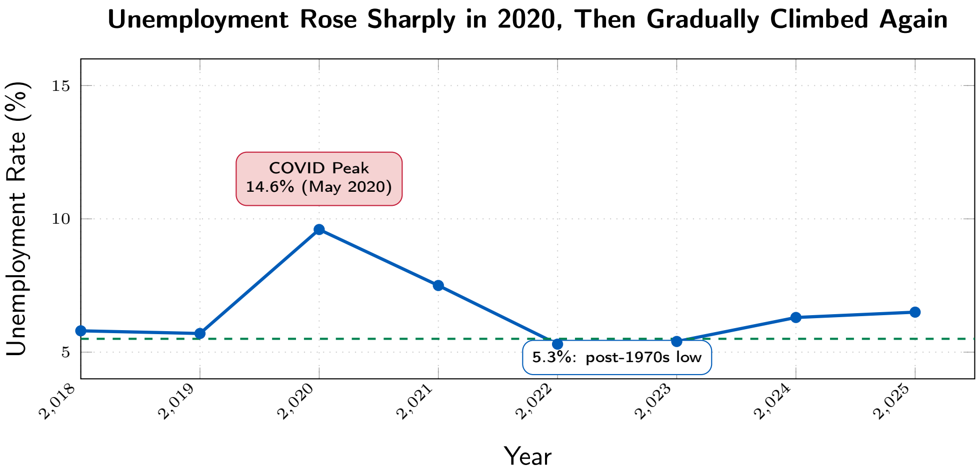


Source: Statistics Canada, Labour Force Survey (Table 14-10-0017-01). 2025 is estimated.

# The Effect of the Business Cycle on Unemployment

- **During recessions:**
  - Firms see sales fall, reduce production, lay off workers.
  - Unemployment often **continues to rise** even after the recession begins (lagging indicator).
  - **Canada 2020**: unemployment hit 14.6% in May 2020.
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  - capital per worker, human capital, and technological change.
- Canada's real GDP per capita quadrupled since 1961 — mostly due to technology and capital deepening.
- **Potential GDP** and the **output gap** reveal whether resources are over- or under-utilized.
- The **financial system** (banks, markets, intermediaries) channels saving into investment via:
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- **Long-run economic growth** is driven by labour productivity, which depends on:
  - capital per worker, human capital, and technological change.
- Canada's real GDP per capita quadrupled since 1961 — mostly due to technology and capital deepening.
- **Potential GDP** and the **output gap** reveal whether resources are over- or under-utilized.
- The **financial system** (banks, markets, intermediaries) channels saving into investment via:
  - risk-sharing, liquidity, and information.
- In a closed economy, $S = I$; government deficits can **crowd out** private investment.
- The **business cycle** — expansions, peaks, recessions, troughs — is accompanied by:
  - higher inflation in booms, lower in recessions.
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# Thank you!

Questions?